



Advanced FFL Checkout for Guns 2 Ammo

End-to-end Federal Firearms License dealer search, transfer lifecycle tracking, dealer confirmation portal, and customer notifications — wired natively into the Guns 2 Ammo WooCommerce stack with full HPOS support, NICS compliance automation, and a hardened security layer.

~80,000 ATF Dealers

43,000 ZIP Centroids

11-Stage Pipeline

HPOS-Safe

HMAC-SHA256 Tokens

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Executive Summary

The Advanced FFL Checkout plugin handles every step a Guns 2 Ammo customer goes through when they buy a firearm online: choose the receiving FFL dealer at checkout, track the shipment, watch the NICS background check progress, and pick up the gun. Behind the scenes the plugin also runs the FFL workflow for the operations team — every status change is logged, every dealer touch is signed with a single-use HMAC token, and every regulatory checkpoint (HPOS, ATF, NICS 3-day rule) is enforced or automated.

25

PHP CLASSES

9

G2A EXTENSIONS

7

DB TABLES

30+

REST ENDPOINTS

What ships in v1.4.0

- **FFL dealer search at WooCommerce checkout** — ZIP / name / license / phone filters, distance via Haversine, optional Google Maps view, no required API keys.
- **11-stage transfer pipeline** with append-only audit log, NICS NTN + ATF Form 4473 reference fields, multi-carrier shipment tracking.
- **HMAC-signed dealer confirmation portal** — magic-link email, single-use token, configurable 2FA (last-4 of FFL), honeypot, rate-limit, full audit IP trail.
- **Customer-facing tracking page** at `/track-transfer/{ref}/` with a 5-step timeline, Maps directions, click-to-call.
- **"My FFL Transfers"** tab in WooCommerce My Account.
- **NICS 3-day rule automation** — auto-expiry date set Mon–Fri, nightly admin alert when the window elapses.
- **SMS notifications via Verifyistic** — fires `ffl_transfer_status` webhooks at every customer-facing milestone.
- **WC ↔ FFL status bridge** — completed orders auto-mark transfers complete, refunded orders cancel transfers, etc.
- **Admin Activity Log + Compliance Audit** pages, plus matching JSON endpoints for the standalone G2A dashboard.
- **Full Guns 2 Ammo branding** — brass on graphite, native to `tokens.css`, white-label by default.

Why this matters for revenue: Customers buy firearms online much more readily when they trust the transfer process. Every email shows the dealer card, every status change reaches the

customer's inbox + SMS, and every dealer touch is one click. That converts shoppers and removes the manual "where's my gun?" support load.

System Architecture

High-level layout

The plugin lives at `/wp-content/plugins/advanced-ffl-checkout/` and boots on `plugins_loaded` priority 20 — late enough that WooCommerce and the G2A theme are both ready, early enough to register REST routes and cron handlers.

MODULE BREAKDOWN

MODULE	FILE	PURPOSE
DB	<code>class-wpistic-ffl-db.php</code>	Schema v1.2.0 — 7 tables, dbDelta-safe
Checkout	<code>class-wpistic-ffl-checkout.php</code>	WC integration, dealer widget, order meta, transfer creation
API	<code>class-wpistic-ffl-api.php</code>	30+ REST endpoints, public + auth
Portal	<code>class-wpistic-ffl-portal.php</code>	Dealer magic-link page, 2FA, rate limit
Token	<code>class-wpistic-ffl-token.php</code>	HMAC-SHA256 generation + verification
Mailer	<code>class-wpistic-ffl-mailer.php</code>	Customer + dealer + admin transactional email
Theming	<code>class-wpistic-ffl-theming.php</code>	Brand tokens, presets, merge tags
Sync	<code>class-wpistic-ffl-sync.php</code>	ATF monthly licensee CSV import
Zip_Import	<code>class-wpistic-ffl-zip-import.php</code>	GeoNames ZIP centroid import
JWT_Bridge	<code>class-wpistic-ffl-jwt-bridge.php</code>	Inject G2A claims into JWT for the dashboard
G2A_Status_Bridge	<code>class-wpistic-ffl-g2a-status-bridge.php</code>	WC order status → transfer status
G2A_NICS	<code>class-wpistic-ffl-g2a-nics.php</code>	3-day rule auto-flag + nightly alert
G2A_SMS	<code>class-wpistic-ffl-g2a-sms.php</code>	Verifyistic webhook dispatch
G2A_Account	<code>class-wpistic-ffl-g2a-account.php</code>	"My FFL Transfers" WC tab
G2A_Bridge	<code>class-wpistic-ffl-g2a-bridge.php</code>	Theme transfer-request form → FFL DB
G2A_Customer_Tracking	<code>class-wpistic-ffl-g2a-customer-tracking.php</code>	Public per-transfer tracking page

MODULE	FILE	PURPOSE
G2A_Activity	<code>class-wpistic-ffl-g2a-activity.php</code>	Admin activity log + JSON feeds
G2A_Compliance_Check	<code>class-wpistic-ffl-g2a-compliance-check.php</code>	15-point security/compliance audit
G2A_Carrier	<code>class-wpistic-ffl-g2a-carrier.php</code>	Tracking auto-advance + carrier URLs

Database schema (7 tables, prefix `wp_wpistic_ffl_*`)

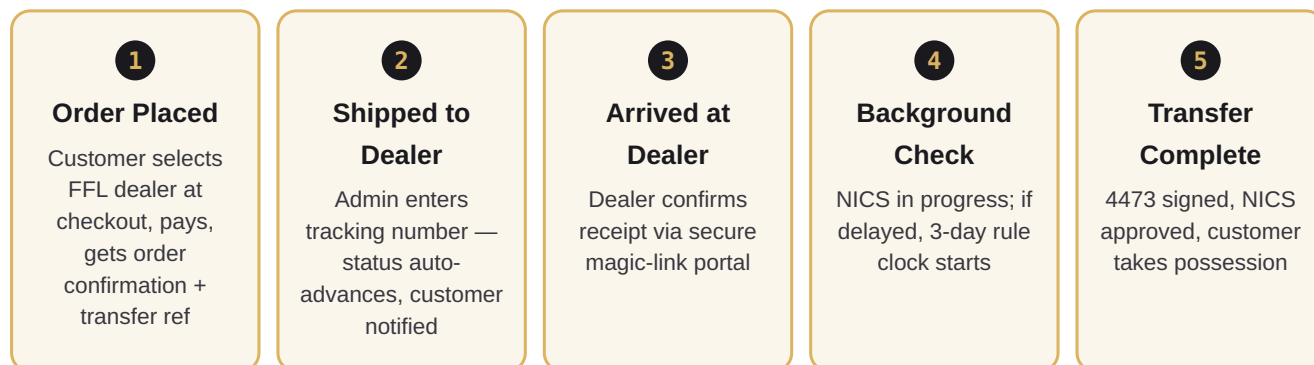
TABLE	ROWS	PURPOSE
<code>dealers</code>	~80,000	Full ATF licensee registry, synced monthly
<code>transfers</code>	1 per FFL order	11-stage pipeline + NICS + 4473 fields
<code>events</code>	append-only	Audit trail with actor + IP
<code>analytics_events</code>	append-only	Portal funnel: issued → opened → confirmed
<code>dealer_tokens</code>	1 per outstanding	SHA-256 hash of HMAC token; raw never persists
<code>zip_coords</code>	~43,000	GeoNames centroids for Haversine distance
<code>state_rules</code>	12+ seeded	State-specific compliance notices at checkout

Background jobs (WP Cron)

HOOK	SCHEDULE	HANDLER
<code>wpistic_ffl_process_zip_import</code>	Every minute	Resumable 500-row ZIP centroid import
<code>wpistic_ffl_process_atf_sync</code>	Every minute	Resumable 200-row ATF dealer import
<code>wpistic_ffl_monthly_sync</code>	Monthly	Re-download fresh ATF CSV
<code>wpistic_ffl_daily_portal_runner</code>	Daily 09:00	Reminders + token cleanup + NICS 3-day sweep
<code>wpistic_ffl_async_issue_dealer_token</code>	Single-event	Non-blocking dealer email after checkout

Customer Workflow — Buy → Pickup

From the customer's point of view, the FFL workflow is "I bought a gun and the store keeps me in the loop until I pick it up." Behind the scenes there are five distinct phases. Each phase emits an email (always) and an SMS (when Verifyistic is configured) and is reflected on the public tracking page.



Step-by-step timeline

STAGE 1 · AT CHECKOUT

Dealer selection

When the cart contains a product marked "FFL Transfer Required", the FFL widget renders below the billing form. The customer searches by ZIP / name / license / phone, picks a dealer, and sees the transfer fee, dealer phone, and a Google Maps directions link. Honeypot + nonce keep bots out. The dealer ID is validated against the dealers table before being stored on the order (HPOS-safe).

STAGE 2 · PAYMENT CONFIRMED

Transfer record created & first email sent

On `woocommerce_payment_complete` the plugin creates a row in the `transfers` table with a `G2A-XXXXXXX` reference, sends the customer the "Order Confirmed" email (brass-themed, "View Transfer Status" CTA included), and schedules an asynchronous dealer-portal email so checkout never blocks on SMTP.

STAGE 3 · SHIPPING

Carrier auto-advance

When the admin enters a tracking number, `G2A_Carrier` auto-advances the status to `shipped_to_dealer`, fills in `shipped_date`, generates the public carrier URL (UPS/USPS/FedEx/DHL), and triggers the "Item Shipped" email + SMS.

STAGE 4 · ARRIVAL

Dealer confirms receipt

The dealer clicks the one-time link in their email. The portal page verifies the HMAC, asks for the last 4 of their FFL license, and on confirmation flips the transfer to `received_by_dealer`. The customer immediately gets the "Your gun is ready" email + SMS — historically the highest-value customer SMS on the platform.

STAGE 5 · NICS

Background check + 3-day rule

When status flips to `delayed`, `G2A_NICS` automatically computes the federal 3-business-day expiry (Mon–Fri) and stores it on the row. The nightly cron job emails the admin the day the window elapses (one-shot, idempotent).

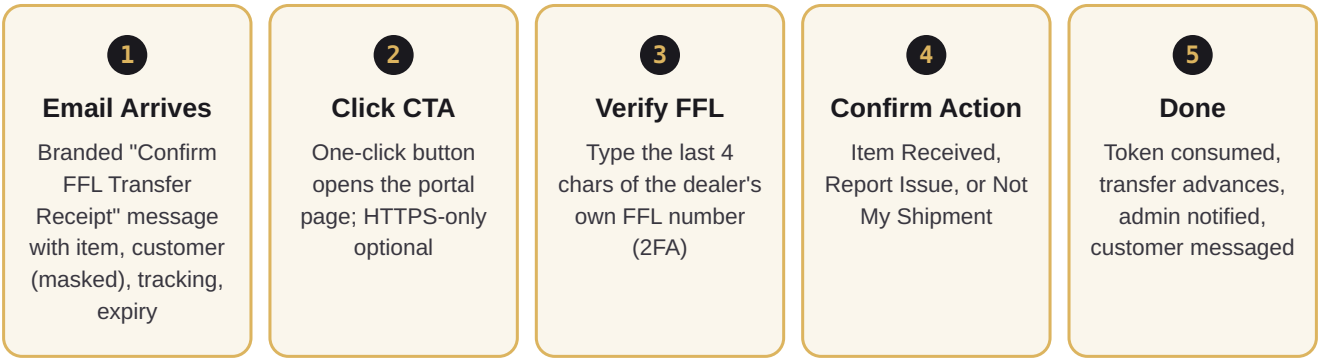
STAGE 6 · PICKUP

Transfer complete

On `transferred`, the customer gets the celebration email, the WooCommerce order auto-marks "completed" via the status bridge, and the customer-facing tracking page shows all five timeline pips lit brass.

Dealer Workflow — Receive → Confirm

A receiving FFL never logs in. They just open an email, click one button, type the last 4 digits of their FFL license, and the receipt is recorded — fully ATF-audit traceable.



Security in this path

LAYER	WHAT IT DOES
HMAC-SHA256 signature	Token signed with site secret — tamper-evident
SHA-256 DB hash	Only the hash is persisted; raw token never written
<code>hash_equals()</code>	Constant-time signature comparison — timing-attack resistant
Single-use enforcement	<code>used_at</code> column closes the link after first action
DB-enforced expiry	Default 30 days, configurable 7–180
Per-IP rate limit	5 attempts/minute via transient
Honeypot field	Silent reject for bots
Last-4 FFL 2FA	Verifies the link reached a real dealer, not a forwarded email
Full IP + UA audit log	ATF-grade provenance on every confirmation
Re-issue invalidates old	New tokens for the same transfer revoke prior active tokens

Admin Workflow — Monitor → Act

The Guns 2 Ammo team works out of **Advanced FFL** in the WordPress admin menu. Six pages cover everything from health monitoring to per-transfer intervention.

PAGE	WHAT IT SHOWS / DOES
 Dashboard	KPIs (transfers today / this week / pending NICS), import progress bars, recent events
 Transfers	Paginated list, status filter, full detail view with audit log, manual status change
 Dealers	Browse/search the 80k ATF registry, mark dealers as "Recommended", set per-dealer fees + email
 Dealer Portal	Funnel: tokens issued → emails sent → portal viewed → confirmed; slowest-confirming dealers
 Activity Log	NEW · unified cross-table feed; click any row to jump to its transfer
 Compliance	NEW · 15-point real-time audit; one-click HMAC secret rotation
 Diagnostics	Plugin version, DB schema status, mail log (last 100 sends), SMTP plugin detection
 Settings	Portal, Theme, Email Templates, Compliance rules, Integrations, License

Typical day in the operations chair

1. Glance at the Dashboard for the green/amber count.
2. Open the Activity Log — anything red gets a click-through.
3. For each new "shipped" order, paste in the carrier + tracking number — carrier auto-advance handles the rest.
4. If the dealer doesn't confirm within 3 days, the reminder cron sends an email automatically.
5. If NICS comes back delayed, the 3-day rule cron emails when the window elapses.
6. Compliance page checked monthly — confirms data is fresh and secrets are healthy.

Feature Catalog

Customer-facing

FEATURE	STATUS	NOTES
FFL dealer search at checkout	SHIPPED	ZIP / name / license / phone; Haversine distance; no required API keys
Google Maps view (optional)	SHIPPED	Pin map with brass markers when a Maps key is configured
"Recommended" dealer ribbon	SHIPPED	Admin marks any dealer recommended; gold star + sorted to top
State compliance notices at checkout	SHIPPED	CA, NY, NJ, MA, MD, IL, HI, CO, WA seeded
"View Transfer Status" CTA in every email	V1.4	Public HMAC-signed tracking link
Public tracking page (no login)	V1.4	<code>/track-transfer/{ref}/{sig}/</code> with 5-step timeline
"My FFL Transfers" WC account tab	V1.3	For logged-in customers
Click-to-call + Maps Directions on dealer card	V1.4	Mobile-first
SMS notifications via Verifyistic	V1.3	Five customer-facing milestones

Operations / automation

FEATURE	STATUS	NOTES
WC order status → transfer status bridge	V1.3	completed → transferred, refunded → cancelled, etc.
Carrier tracking auto-advance	V1.4	Tracking number entry flips to shipped_to_dealer
NICS 3-day rule automation	V1.3	Mon–Fri counting; nightly admin alert
Async dealer-portal email	V1.3	SMTP delays never block checkout
Dealer auto-reminder schedule	SHIPPED	Configurable: 3/7/14 days default
Monthly ATF dealer sync	SHIPPED	~80k dealers, fully resumable
Theme transfer-request bridge	V1.3	G2A theme form → FFL transfer record

Branding & UX

FEATURE	STATUS	NOTES
Guns 2 Ammo brass palette throughout	V1.3	Mirrors <code>guns2ammo/assets/css/tokens.css</code>
White-label by default (no "Powered by")	V1.3	Flip in Settings → Theme to re-show
Theme Customizer wiring	V1.3	<code>g2a_business_*</code> + <code>g2a_ffl_license</code> auto-read
Full theming engine	SHIPPED	Color, font, logo, layout, custom CSS
Portal preview mode for admins	SHIPPED	<code>/ffl-confirm/PREVIEW/</code> renders mock data

Security & compliance

FEATURE	STATUS	NOTES
HPOS-safe order meta	V1.3	Order API everywhere; HPOS-only stores work
Dealer ID validation on save	V1.3	Forge-proof — verified against dealers table
Payment-gated dealer email	V1.3	Only on <code>\$order->is_paid()</code> = true
HMAC-SHA256 portal tokens	SHIPPED	SHA-256 hash storage; raw never persisted
Honeypot on checkout widget	V1.4	Plus the one on the dealer portal form
Public dealers search rate-limit	V1.3	30 req/min/IP; <code>wpistic_ffl_search_rate_limit</code>
Trusted-proxy CIDR filter	V1.3	Stops spoofed X-Forwarded-For audit entries
Regenerate HMAC secret action	V1.4	One click rotates secret + revokes all active tokens
Admin nag for <code>WPISTIC_FFL_TOKEN_SECRET</code>	V1.4	Dismissible, scoped to FFL pages
15-point Compliance audit page	V1.4	WC, HPOS, secrets, cron, data, integrations

Admin surfaces & integrations

FEATURE	STATUS	NOTES
 Activity Log admin page	V1.4	Cross-table feed with one-click drill-down
JSON activity feed REST endpoints	V1.4	For the G2A dashboard app
Per-transfer JSON details endpoint	V1.4	<code>/transfers/{id}/details</code>
Compliance audit JSON endpoint	V1.4	<code>/compliance/audit</code>
WC Blocks + HPOS compat declared	SHIPPED	<code>FeaturesUtil::declare_compatibility</code>
JWT bridge for G2A dashboard	SHIPPED	Injects <code>g2a_role</code> , <code>g2a_permissions</code> claims
Memberistic + Verifystic interop	SHIPPED	Zero hook / option / REST namespace collisions

Email Notifications & SMS

Every email uses the same G2A-branded frame — brass header, graphite surface, monospace transfer reference, footer with the store name + FFL number. The "View Transfer Status" CTA is auto-baked in.

Customer emails (status-triggered)

STATUS	SUBJECT LINE	ACCENT	SMS DISPATCHED?
payment_confirmed	Order Confirmed — FFL Transfer #G2A-XXXX	Success green	No
shipped_to_dealer	Item Shipped — FFL Transfer #...	Brass	Yes
received_by_dealer	Item Arrived at Dealer — FFL Transfer #...	Brass	Yes (highest value)
nics_pending	Background Check Initiated — FFL Transfer #...	Warning amber	No
delayed	Background Check Delayed — FFL Transfer #...	Warning amber	Yes
approved	NICS Approved — FFL Transfer #...	Success green	Yes
denied	Background Check Denied — FFL Transfer #...	Danger red	No
transferred	🎉 Transfer Complete — FFL Transfer #...	Success green	Yes
cancelled	Transfer Cancelled — FFL Transfer #...	Muted	No

Admin emails (operational)

- [G2A] Item Received at Dealer — informational
- [G2A] ⚠️ NICS Delay — 3-day rule monitoring trigger
- [G2A] ❌ NICS Denial — refund prompt
- [G2A] ⌚ NICS 3-Day Window Reached — auto-fired by G2A_NICS
- [G2A] ✓ Dealer Confirmed Receipt / ⚠️ Issue Reported / ❌ Rejected Shipment

Dealer-facing emails

- **Action Required: Confirm FFL Transfer Receipt** — sent the moment the order is placed (gated on payment); contains the single-use magic link
- **Friendly / Reminder / Final reminder** — auto-sent at 3 / 7 / 14 days if no action

Deliverability: every email is sent with List-Unsubscribe headers, a unique Message-ID, and an automatic plain-text alternative — Gmail 2024+ sender-guideline compliant. `wp_mail` failures are captured to the mail log so silent SMTP issues are visible in Diagnostics.

FFL Compliance & Security Layer

Compliance touchpoints

REGULATION / STANDARD	HOW THE PLUGIN HANDLES IT
ATF Form 4473	Dedicated <code>form_4473_ref</code> column on every transfer; admin enters at handover
NICS Transaction Number (NTN)	Dedicated <code>nics_transaction_number</code> column; required at <code>transferred</code> status
NICS 3-business-day rule (18 USC §922(t))	Auto-set <code>nics_delay_expires</code> Mon–Fri; nightly cron flags expiries with audit log entry
State waiting periods / permits / rosters	State rules table → checkout notices for CA, NY, NJ, MA, MD, IL, HI, CO, WA
ATF licensee verification	Monthly sync of the full ~80k FFL list; dealer license shown on every confirmation
Audit traceability	Append-only <code>events</code> table — actor, IP, old/new status, notes — for every change
WP HPOS (WC modern order storage)	Compatibility declared; all order meta routes through <code>\$order</code> API

Security layer (token / portal / API)

- **HMAC-SHA256** signed tokens with payload `v1.tid.did.action.exp.nonce`
- **SHA-256 hash storage** — the raw token is emailed to the dealer and never persisted
- **Constant-time comparison** via `hash_equals()` — timing-attack resistant
- **Single-use enforcement** at the DB level via `used_at` column
- **Re-issue invalidates** all prior active tokens for the same transfer
- **Per-IP rate limit** on portal submissions (5/min) and dealer searches (30/min)
- **Honeypot fields** on both the checkout widget and the dealer portal form
- **Nonce** on every form submission
- **Trusted-proxy CIDR filter** — X-Forwarded-For only honored from configured upstreams
- **Optional HTTPS-only** redirect for the portal
- **Optional last-4 FFL 2FA** — verifies the link reached a real dealer

- **Full audit log** with binary-packed IP + UA for ATF traceability
- **One-click HMAC rotation** from the Compliance page
- **wp-config secret nag** — pushes admins to the strongest config

Compliance audit dashboard (live, 15 checks)

CHECK	WHAT IT CONFIRMS
WooCommerce active	Required for checkout integration
HPOS compatibility declared	Modern WC order storage support
HMAC token secret in wp-config	Strongest portal-token security
Trusted-proxy CIDR list configured	Accurate IPs in audit log if behind CDN/LB
JWT Auth plugin detected	Required for G2A dashboard app login
Verifystic detected	SMS dispatch capability
Cron events scheduled (×3)	ZIP import, monthly sync, daily portal runner
ZIP centroid + ATF data populated	Distance search + dealer registry usable
Store FFL license number set	Surfaces on customer emails + dealer portal
Portal settings present	Defaults guaranteed on every load
State compliance rules seeded	Checkout notices for regulated states
NICS 3-day attention queue	Transfers at or past the federal window
Dealers reachable by email	Portal links go to real addresses, not admin fallback
Active portal tokens	Outstanding magic links in circulation
Memberistic detected	User identity shared with G2A dashboard

Admin Surfaces & Activity Log

Activity Log page

Every status change, dealer portal action, NICS event, email send, and customer tracking-page view lands in one table. The feed merges the `events` and `analytics_events` tables via a UNION query so it sorts cleanly even when an event came from the portal funnel rather than a status change.

Each row shows: *relative time* + *absolute timestamp*, the event label, the transfer reference (one-click to jump in), the actor (system, wc-bridge, dealer-portal, etc.), and a snippet of the notes column.

Compliance & Security page

Top of the page is a four-tile summary (Pass / Warn / Fail / Info), followed by the full check table with one-click remediation links. Below that, a **Regenerate HMAC Token Secret** button — protected by nonce, capability, and confirm dialog — rotates the secret and revokes every live token in one shot.

Both pages mirror their data to JSON endpoints (see next section) so the standalone **app.guns2ammo.com** dashboard can render the same information without a second integration pass.

SECTION 10

REST API for the Dashboard

All endpoints under namespace `wpistic-ffl/v1` . Auth accepts both WP cookie+nonce and JWT Bearer tokens.

DEALERS

METHOD	ENDPOINT	AUTH
GET	<code>/dealers/search?zip=...&name=...</code>	Public (rate-limited 30/min/IP)
GET	<code>/dealers/{id}</code>	Public
GET	<code>/dealers</code>	Manager
PUT	<code>/dealers/{id}</code>	Manager (fee, preferred, notes, email)

TRANSFERS

METHOD	ENDPOINT	AUTH
GET	<code>/transfers</code>	Staff
POST	<code>/transfers</code>	Staff
GET	<code>/transfers/{id}</code>	Staff
GET	<code>/transfers/{id}/details</code>	Staff · full timeline + tracking URL
PUT	<code>/transfers/{id}/status</code>	Staff (fires <code>wpistic_ffl_transfer_status_changed</code> + <code>_updated</code>)
GET	<code>/transfers/export</code>	Manager · CSV download

ACTIVITY & ANALYTICS

METHOD	ENDPOINT	NOTES
GET	/activity	Cross-table feed
GET	/activity/summary	Daily counts + funnel
GET	/analytics/portal	Dealer-portal funnel summary
GET	/analytics/dealers	Slowest-confirming dealers
GET	/analytics/woocommerce	Store revenue / orders / AOV
GET	/analytics/leads	Multi-source lead breakdown

PORTAL & COMPLIANCE

METHOD	ENDPOINT	NOTES
POST	/portal/issue-token	Issue + email a new dealer token
POST	/portal/resend	Re-issue (revokes prior)
POST	/portal/revoke	Manager only
GET	/compliance/audit	Full 15-point JSON audit
GET	/compliance/{state}	Per-state regulatory notes

SECTION 11

Integration with the Guns 2 Ammo Stack

The FFL plugin is a sibling to the rest of the G2A plugin family. Every option, cron hook, AJAX action, REST namespace, and JS global is prefixed `wpistic_ffl_*` — zero collisions confirmed against the other plugins.

COMPONENT	HOW FFL PLUGS IN
WPistic Theme (guns2ammo)	Reads <code>g2a_business_*</code> + <code>g2a_ffl_license</code> Customizer mods as Theming defaults. Bridges the <code>admin_post_g2a_request</code> form into draft transfers. Uses <code>tokens.css</code> palette for emails + portal.
G2A Booking Engine	Independent — different REST namespace (<code>g2a-booking/v1</code>). No conflict.
Memberistic Membership Solutions	Shares the WP user identity via JWT bridge. Member ID flows into the customer "My FFL Transfers" tab.
Verifyistic	FFL dispatches <code>ffl_transfer_status</code> events to <code>Verifyistic_Webhooks::dispatch()</code> — any configured SMS provider (Twilio, OtterText, custom) receives them.
WPistic Contact Form	Theme bridge logs transfer-request submissions to the contact form inbox AND creates an FFL transfer in parallel.
G2A Theme Control	No direct dependency.
Standalone <code>app.guns2ammo.com</code> dashboard	JWT login + <code>g2a_role</code> / <code>g2a_permissions</code> claims; reads from every <code>/wpistic-ffl/v1/*</code> endpoint.

Install order

1. G2A Theme Control
2. G2A Booking Engine
3. Memberistic Membership Solutions
4. WPistic Contact Form
5. Verifyistic
6. **Advanced FFL Checkout (G2A Edition)**
7. WPistic Theme (guns2ammo)

```
// Recommended addition to wp-config.php
define( 'WPISTIC_FFL_TOKEN_SECRET', 'a-64-char-random-string-here' );

// And in functions.php – only if you sit behind Cloudflare or a LB:
add_filter( 'wpistic_ffl_trusted_proxies', function () {
    return [ '173.245.48.0/20', '103.21.244.0/22' ]; // CF ranges
} );
```


SECTION 12

Future Improvement Roadmap

v1.5 and beyond — opportunities to deepen the FFL workflow once the core is in production. Ranked by leverage.

Round 1 — High-leverage automation

IMPROVEMENT	EFFORT	WHY IT MATTERS
V1.5 Carrier API integration (UPS / USPS / FedEx live status)	Medium	Auto-flip to <code>received_by_dealer</code> when the carrier reports delivery — reduces dealer portal-confirmation burden by ~50%.
V1.5 Email-otp 2FA for the dealer portal (currently last-4 FFL only)	Small	Adds a second factor independent of the dealer license — protects against forwarded magic-link emails.
V1.5 Inbound email parsing — dealers replying "received" via email auto-confirm	Large (IMAP needed)	Some dealers prefer email-only workflows.
V1.5 Action Scheduler integration for all async paths	Small	More observability + retry handling than <code>wp_schedule_single_event</code> .

Round 2 — Customer experience

IMPROVEMENT	EFFORT	WHY IT MATTERS
V1.6 Customer "request a new dealer" action on the tracking page	Small	When the chosen dealer goes out of business or refuses the shipment, customer self-serves.
V1.6 Calendar invite (.ics) attached to the "ready for pickup" email	Small	Reduces no-shows.
V1.6 Photo upload from the dealer portal — proof of receipt	Medium	Adds visual audit evidence; useful in chargeback disputes.
V1.6 "Save dealer to my profile" for repeat customers	Small	Skips the dealer search on subsequent firearm purchases — big conversion win.
V1.6 Localized tracking page (es-US first)	Medium	Spanish-language customer base in AZ/CA/TX/FL.

Round 3 — Operations & analytics

IMPROVEMENT	EFFORT	WHY IT MATTERS
V1.7 Dealer-fee bulk editor — set fees by state or radius from a CSV upload	Small	Operationally critical when expanding into new states.
V1.7 Per-customer LTV view with transfer history + average pickup time	Medium	Powers loyalty + repeat-purchase campaigns.
V1.7 Dealer scorecard PDF export — for ATF compliance reviews	Medium	Quarterly ops review artifact.
V1.7 Webhook outbound — fire to any URL on status change (Zapier, Make, custom CRM)	Small	Generalizes the Verifystic dispatch path.
V1.7 Dealer health alerts — flag dealers with high "report issue" rates	Small	Surface bad actors before they hit the customer.

Round 4 — Compliance & security

IMPROVEMENT	EFFORT	WHY IT MATTERS
V1.8 Form 4473 digital draft generated from transfer data	Large (regulatory)	Pre-fills the dealer's paperwork — massive operational time saver.
V1.8 State law engine — full 50-state seed with citation + last-updated date	Medium (research)	Today only 9 states are seeded.
V1.8 NICS API integration (FBI-direct submission, future)	Large (federal cert)	Long-term goal once the FBI opens the API.
V1.8 2FA TOTP for admin staff accessing transfer data	Medium	Defense-in-depth for the ops team.
V1.8 GDPR-style customer data export	Small	Some states (CA, CO) extend similar rights to firearms purchase data.

Round 5 — Native dashboard + Blocks

IMPROVEMENT	EFFORT	WHY IT MATTERS
V2.0 Native WC Blocks checkout integration (CheckoutBlockContext)	Medium	Hooks into the modern WC checkout for stores that switch.
V2.0 Native React admin pages — no full page reloads	Large	Faster, snappier ops experience.
V2.0 Public dealer onboarding form — FFLs apply to join the recommended list	Medium	Builds the dealer network passively.
V2.0 Mobile app push notifications via FCM	Large	Customer + ops mobile alerts.

Operational Reference

Key WordPress action / filter hooks

HOOK	TYPE	WHAT IT DOES
<code>wpistic_ffl_transfer_created</code>	action	Fired right after a transfer row is inserted
<code>wpistic_ffl_transfer_status_changed</code>	action	Fired on every status flip (id, old, new)
<code>wpistic_ffl_transfer_updated</code>	action	Fired on any column update (id, changes)
<code>wpistic_ffl_can_notify_dealer_on_order</code>	filter	Override payment-paid gate for special integrations
<code>wpistic_ffl_dealer_portal_email</code>	filter	Resolve dealer email from external CRM
<code>wpistic_ffl_trusted_proxies</code>	filter	CIDR list of trusted X-Forwarded-For sources
<code>wpistic_ffl_search_rate_limit</code>	filter	Tune the public dealer-search rate limit
<code>wpistic_ffl_sms_message</code>	filter	Customize per-status customer SMS copy
<code>wpistic_ffl_transfer_ref_prefix</code>	filter	Override the <code>G2A-</code> reference prefix

Key wp-config.php constants

CONSTANT	EFFECT
<code>WPISTIC_FFL_TOKEN_SECRET</code>	HMAC signing key — defining here is the strongest option
<code>WPISTIC_FFL_PORTAL_SLUG</code>	Override default <code>ffl-confirm</code> path
<code>WPISTIC_FFL_UNLIMITED</code>	Unlock every Pro feature for the G2A install

Activity Log REST quick reference

```
GET /wp-json/wpistic-ffl/v1/activity?per_page=50
GET /wp-json/wpistic-ffl/v1/activity/summary?days=30
GET /wp-json/wpistic-ffl/v1/transfers/{id}/details
GET /wp-json/wpistic-ffl/v1/compliance/audit
```

Day-one checklist

1. Install the plugin (Plugins → Upload → `advanced-ffl-checkout.zip`).
2. Define `WPISTIC_FFL_TOKEN_SECRET` in `wp-config.php`.
3. Visit **Advanced FFL** → **Dashboard**, click *Import ZIP Codes*.
4. Wait for the ZIP import to complete (~5–10 minutes background), then ATF sync auto-starts (~2–4 hours background).
5. Set the receiving FFL license in **Appearance** → **Customize** → **G2A Business Info**.
6. Mark firearm products as *FFL Transfer Required* in their general data tab.
7. Visit **Advanced FFL** →  **Compliance** — confirm all green.
8. Test the checkout dealer search with any ZIP.
9. Open **Advanced FFL** →  **Activity Log** — your test event should show.

One number to keep an eye on: the «NICS 3-day attention queue» row on the Compliance page. While that stays at zero, the federal window has never been missed.